Unit 1

Lesson Two

Finding the term containing x^r in the expansion of binomial

To find the term containing x^K in the expansion of $(x + a)^n$ according to the descending exponents of x , follow the following :

1. Find the term T_{r+1} in its simplest form to be able to determine the power of the variable x in terms of r
2. Equate the resultant power of x in the term T_{r+1} and the required power K to get the value of r and hence determine the term which contains x^K and it is T_{r+1}
3. Find the term which contains x^K by substituting by the value of r that we got in T_{r+1}

Remarks

1. If the value of r that we got doesn't belong to the set of natural numbers , then there is no term containing the required x^K in this expansion.
2. To find the term free of x , we consider that we need to get the term contains x^0
   i.e. Put the power of x in T_{r+1} equal zero , then get the value of r

Example 1

In the expansion of : $\left(2x^2 + \frac{1}{x}\right)^{10}$ according to the descending exponents of x , find the term which contains x^8

Solution

Let the term contains x^8 be T_{r+1}

, $\because T_{r+1} = {}^{10}C_r \left(\frac{1}{x}\right)^r (2x^2)^{10-r} = {}^{10}C_r \times x^{-r} \times 2^{10-r} \times x^{20-2r} = {}^{10}C_r \times 2^{10-r} \times x^{20-3r}$

$\therefore$ put $20 - 3r = 8$ $\quad\quad$ $\therefore 3r = 12$ $\quad\quad$ $\therefore r = 4$

$\therefore T_5$ is the term which contains x^8

$\therefore T_5 = {}^{10}C_4 \times 2^{10-4} \times x^{20-3(4)} = 13440\, x^8$

Example 2

Find the term free of x in the expansion of : $\left(x+\frac{1}{4x^3}\right)^{16}$ according to the descending exponents of x

Solution

$\because T_{r+1} = {}^{16}C_r \left(\frac{1}{4x^3}\right)^r (x)^{16-r} = {}^{16}C_r \times \left(\frac{1}{4}\right)^r \times x^{-3r} \times x^{16-r} = {}^{16}C_r \times \left(\frac{1}{4}\right)^r \times x^{16-4r}$

Put : $16 - 4r = 0$ to find the term containing x^0 (the term free of x)

$\therefore 4r = 16 \qquad \therefore r = 4$

$\therefore T_5 = {}^{16}C_4 \times \left(\frac{1}{4}\right)^4 \times x^0 = 1820 \times \frac{1}{256} = \frac{455}{64}$

Example 3

In the expansion of : $\left(x^2+\frac{1}{x^2}\right)^{12} - \left(x+\frac{1}{x}\right)^{12}$, find the term containing x^4

Solution

Determine the term that contains x^4 in the expansion of each of the two expressions

$\left(x^2+\frac{1}{x^2}\right)^{12}$, $\left(x+\frac{1}{x}\right)^{12}$

First : for the expansion of : $\left(x^2+\frac{1}{x^2}\right)^{12}$:

$T_{r+1} = {}^{12}C_r \left(\frac{1}{x^2}\right)^r (x^2)^{12-r} = {}^{12}C_r \times x^{-2r} \times x^{24-2r} = {}^{12}C_r \times x^{24-4r}$

$\therefore 24 - 4r = 4 \qquad \therefore 4r = 20 \qquad \therefore r = 5$

$\therefore$ The term consists of x^4 is $T_{5+1} = T_6$

$\therefore T_6 = {}^{12}C_5 \, x^4$ (1)

Second : for the expansion of : $\left(x+\frac{1}{x}\right)^{12}$:

$T_{r+1} = {}^{12}C_r \left(\frac{1}{x}\right)^r (x)^{12-r} = {}^{12}C_r \times x^{-r} \times x^{12-r} = {}^{12}C_r \times x^{12-2r}$

$\therefore 12 - 2r = 4 \qquad \therefore 2r = 8 \qquad \therefore r = 4$

$\therefore$ The term contains x^4 is $T_{4+1} = T_5$

$\therefore T_5 = {}^{12}C_4 \, x^4$ (2)

From (1) , (2) :

$\therefore$ The term consists of x^4 in the given expression $= {}^{12}C_5 \, x^4 - {}^{12}C_4 \, x^4 = 297 \, x^4$

Example 4

Find the coefficient of x^{14} in the expansion of : $32\,x^4\left(\frac{x^2}{2}+\frac{2}{x^3}\right)^{15}$ according to the descending exponents of x

Solution

$$T_{r+1} = 32\,x^4 \times {}^{15}C_r \left(\frac{2}{x^3}\right)^r \left(\frac{x^2}{2}\right)^{15-r}$$

$$= 32\,x^4 \times {}^{15}C_r\,(2)^r \left(\frac{1}{2}\right)^{15-r} \times x^{-3r} \times x^{30-2r}$$

$$= 32 \times {}^{15}C_r\,(2)^r \left(\frac{1}{2}\right)^{15-r} \times x^{4-3r+30-2r} = 32 \times {}^{15}C_r\,(2)^r \left(\frac{1}{2}\right)^{15-r} \times x^{34-5r}$$

to get the coefficient of x^{14}

put $34 - 5r = 14$ $\quad\therefore r = 4$

$\therefore$ The coefficient of $x^{14} = 32\ {}^{15}C_4\,(2)^4 \left(\frac{1}{2}\right)^{11} = \frac{1365}{4}$

Another solution

In the expansion of : $32\,x^4\left(\frac{x^2}{2}+\frac{2}{x^3}\right)^{15}$

the term contains x^{14} equals $32\,x^4 \times$ the term contains x^{10} so we can get the term contains x^{10} in the expansion of : $\left(\frac{x^2}{2}+\frac{2}{x^3}\right)^{15}$ and it is assumed to be T_{r+1} where :

$$T_{r+1} = {}^{15}C_r \left(\frac{2}{x^3}\right)^r \left(\frac{x^2}{2}\right)^{15-r} = {}^{15}C_r\,(2)^r \left(\frac{1}{2}\right)^{15-r} \times x^{-3r} \times x^{30-2r}$$

$$\therefore T_{r+1} = {}^{15}C_r\,(2)^r \left(\frac{1}{2}\right)^{15-r} \times x^{30-5r}$$

$\therefore 30 - 5r = 10$ $\quad\therefore r = 4$

$\therefore$ Coefficient of $x^{10} = {}^{15}C_4\,(2)^4 \left(\frac{1}{2}\right)^{11}$

$\therefore$ The coefficient of $x^{14} = 32 \times {}^{15}C_4\,(2)^4 \left(\frac{1}{2}\right)^{11} = \frac{1365}{4}$

Example 5

Prove that the expansion of : $\left(x^2+\frac{2}{x^3}\right)^{11}$ has no term free of x or contains x^3 but it has a term contains x^{-3}

Solution

$$\because T_{r+1} = {}^{11}C_r \left(\frac{2}{x^3}\right)^r (x^2)^{11-r} = {}^{11}C_r\,(2)^r\,x^{-3r} \times x^{22-2r}$$

$$\therefore T_{r+1} = {}^{11}C_r\,(2)^r \times x^{22-5r}$$

If the expansion contains a term free of x , then $22 - 5r = 0$

$\therefore 5r = 22$ $\therefore r = 4\frac{2}{5}$

, $\because$ r must be a natural number

$\therefore$ There is no term free of x in this expansion

and if the expansion contains a term contains x^3 , then : $22 - 5r = 3$

$\therefore 5r = 19$ $\therefore r = 3\frac{4}{5}$

, $\because$ r must be a natural number

$\therefore$ There is no term contains x^3 in this expansion

and if the expansion contains a term contains x^{-3} , then : $22 - 5r = -3$

$\therefore 5r = 25$ $\therefore r = 5$

$\therefore$ The expansion including term contains x^{-3} and it is $T_{5+1} = T_6$

Example 6

If the expansion of : $\left(x^4 + \frac{1}{x^3}\right)^n$ includes a term free of x ,

Prove that : n = 7 or multiple of 7 and find the term free of x when n = 7

Solution

$T_{r+1} = {}^nC_r \left(\frac{1}{x^3}\right)^r (x^4)^{n-r} = {}^nC_r\, x^{-3r} \times x^{4n-4r} = {}^nC_r\, x^{4n-7r}$

, $\because$ The expansion contains a term free of x

$\therefore 4n - 7r = 0$ $\therefore r = \frac{4n}{7}$

, $\because$ r must be a natural number $\therefore$ n has to be divisible by 7

$\therefore$ n = 7 or multiple of 7

when n = 7 , then $r = \frac{4 \times 7}{7} = 4$ thus the term free of x is $T_{4+1} = T_5$

, then $T_5 = {}^7C_4 = 35$

i.e. The term free of $x = 35$

Example 7

In the expansion of : $\left(x^K + \frac{1}{x}\right)^{12}$ where $K \in \mathbb{Z}^+$, find :

1. The values of K that make the expansion has a term free of x

2. The ratio between the term free of x and the coefficient of the middle term for the biggest value of K you got from (1)

Solution

1. Assume that the term free of x is T_{r+1}

$\therefore T_{r+1} = {}^{12}C_r \left(\frac{1}{x}\right)^r (x^K)^{12-r} = {}^{12}C_r \times x^{-r} \times x^{12K-Kr}$

$\therefore T_{r+1} = {}^{12}C_r \times x^{12K-Kr-r}$

to obtain the term free of x

$\therefore 12K - Kr - r = 0$ $\quad\quad$ $\therefore 12K = Kr + r$ $\quad\quad$ $\therefore 12K = r(k+1)$

$\therefore r = \frac{12k}{k+1}$, $\because r \in \mathbb{N}$, $k+1$ can not be a divisor of k

$\therefore k+1$ is a divisor from divisors of 12

$\therefore k+1 = 1$ and so $k = 0$ (Refused because $k \in \mathbb{Z}^+$)

, $k+1 = 2$ and so $k = 1$

, $k+1 = 3$ and so $k = 2$

, $k+1 = 4$ and so $k = 3$

, $k+1 = 6$ and so $k = 5$

, $k+1 = 12$ and so $k = 11$

2. $\because$ The biggest value of $K = 11$, this is when $r = 11$

$\therefore$ The term free of x is $T_{12} = {}^{12}C_{11} = {}^{12}C_1 = 12$

, the middle term $= T_{\frac{12}{2}+1} = T_7$

$\therefore$ The coefficient of the middle term $= {}^{12}C_6 = 924$

$\therefore \frac{\text{The term free of } x}{\text{The coefficient of the middle term}} = \frac{12}{924} = \frac{1}{77}$